CHAEHYUN LIM

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EDUCATION

Georgia Institute of Technology — M.S. Aerospace Engineering, Expected May 2027

Georgia Institute of Technology — B.S. Aerospace Engineering, May 2026 | GPA 3.65/4.00, Highest Honor

ENGINEERING EXPERIENCE & PROJECTS

Moore Dynamics & Analytics Lab, Georgia Tech — Undergraduate Researcher | Oct 2024–May 2026

• Developed MATLAB GUI integrating Arduino-controlled impact actuation, NI cDAQ data acquisition, calibration, visualization, and data logging.

• Implemented camera-pattern recognition and target-coordinate calculation; supported UR30 positioning and resolved driver and MATLAB faults.

• Designed SolidWorks fixtures and optimized SLA printing; project test range approximately 250–1,500 N (coupled) and 3,000 N (released).

43rd VFS Student Design Competition — Propulsion Lead | Aug 2025–Jun 2026

• Developed hybrid-electric power architecture and MATLAB mission-energy models for hover, conversion, and cruise; evaluated turbogenerator/battery power allocation.

• Coordinated propulsion interfaces with aerodynamics and structures; team placed 2nd in undergraduate division.

Reaction-Wheel Heading Control, Georgia Tech — Controls Project | Spring 2026

• Designed and validated PI-II reaction-wheel prototype; integrated Arduino, IMU, ESC, and BLDC hardware.

• Team results: 2.7 s rise, 5.9 s settling, <8.24% overshoot, ±2.8° heading error; ranked #1 in class.

Vertical Lift Research Center of Excellence, Georgia Tech — Battery Test Operator | Jan–May 2025

• Conducted 6S Li-polymer battery cycle-life tests using published protocol (≤1C charge, 750 W constant-power discharge); recorded capacity and resistance data.

UAV System Design & Flight Operations — Pilot and System Designer | Aug 2024–Dec 2025

• Integrated UAV structure, 3D-printed battery enclosure, and thermal/optical sensors; performed waypoint flights to assess stability and payload function.

LEADERSHIP

ROK Defense Intelligence Command — Squad Leader | Oct 2020–Apr 2022

• Led 11-person squad, authored 100-page operations guide, oversaw 33 defense/aerospace publications, and served as interpreter in a U.S.–ROK combined exercise.

SKILLS & CERTIFICATIONS

Test: NI cDAQ, multimeter, power supply.

Tools: MATLAB, Arduino, Excel, SolidWorks, Autodesk Inventor, SLA printing, Python.

UAV: FAA Part 107, EASA A1/A3, ROK UAV qualification (20+ logged flight hours).

Languages: Korean (native), English (fluent), Mandarin Chinese (HSK 5).